

**ADMINISTRATIVE INFORMATION****FMI No: 25358****FMI Title: kV Circuit Board Replacement**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☒ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☒	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☐ Batch with	_____
	To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked:****FMI Instructions Part No.: 5132113-100****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas

GEMS - Europe

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25358

FMI Title: kV Circuit Board Replacement

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

Effectivity

Gantry S/N	SID	Costumer
350540CN1	513584PRO	THE HEALTH ALLIANCE OF GREATER
348625CN5	423738CT01	AUFFERMANN AM BETHESDA KHS
352577CN1	616391RTCT	SPECTRUM HEALTH DOWNTOWN
353176CN1	856210062	CHULALONGKORN HOS
352968CN2	QV0024	TAKASAGO SHIMIN BYOUIN
351405CN6	M2405503	DU PONT DE VAAST
351437CN9	410543LS16	PENINSULA REG MED CTR /SH
351433CN8	713442KCT2	MEDPARTNERS, INC
351402CN3	907562CTPRO	PROVIDENCE IMAGING CENTER
351395CN9	918249ICT	INFINEDI HEALTH S
353489CN8		
351764CN6	00243CTS07	UNIVERSITY HOSPITAL OF WALES
352715CN7	908522PROCT	OVERLOOK HOSPITAL
353034CN2	419793CT01	SCHULTZE,SCHULENBURG,RUEBESAM,
352229CN9		
352945CN0		VOJENSKA NEMOCNICE OLOMOUC
353147CN2	604874CTLS	MOUNT SAINT JOSEPH HOSPITAL
353577CN0	604278CTLS	THE RICHMOND HOSP
353578CN8		
355814CN5	205759CT6	DCH REGIONAL MEDI
353081CN3	623327CT16P	BANNER HEALTH SYSTEM ARIZONA
353652CN1	030013CT04	C.H. DE LUXEMBOURG
353576CN2	82421040131	BJ POLICE GENERAL HOSPITAL
355089CN4	EE1019CT01	PõHJA-EESTI REGIN
354243CN8	QV0040	TAKEDA SOUGOU BYOUIN
354685CN0	FI1025CT01	SEIN#JOEN KESKUSS
355095CN1	828879VSIM	VALDESE GENERAL
354973CN0	82421120184	JIANGSU XUZHOU ME
355430CN0	475150CT01	HOPITAL RIVIERA SITE DE VEVEY
355275CN9		
355094CN4	717267CT16	CHAMBERSBURG HOSPITAL/VHA

354970CN6	606327TSRT	TRI STATE REGIONAL CANCER CTR
354969CN8	82421120181	XU ZHOU #1 PEOP HOSP
354946CN6	210616LSRT	CANCER THERAPY & RES CTR
354972CN2		
355132CN2	2530313	UNIV OF MED & DENT OF NJ
355063CN9		
355133CN0	A5632413	ISTITUTO NEUROMED S.R.L.
355222CN1	605977LSPRO	HEART HOSPITAL OF SOUTH DAKOTA
355223CN9	AE1016CT02	SHEIKH KHALIFA BIN ZAYED (ABU
355789CN9	853210158	TZU-CHI HOSPITAL TAIPEI BRANCH
355790CN7	734712CT2PRO	MISSION HEALTH/ST JOSEPH MERCY
355809CN5		
356021CN6	608324CT1	MONROE CLINIC HOSPITAL
356000CN0	215612LSPRO	FRANKFORD HOSPITAL
356997CN7	916784CT2	KAISER HOSPITAL ASSET MGMT
357543CN8	82421040135	BEIJING CHAOYANG HOSP.
357082CN7		
357486CN0	813615LS16	UNIVERSITY COMMUN
357153CN6	M4164919	BEAUVAIS
356827CN6	82421040136	BEIJING PINGGU HOSPITAL
357849CN9	317582LSPRO	ST VINCENT HOSPITAL
358060CN2	QV0028	ZAIDANHOUSINGANKE
357844CN0	QV0026	SYOUWA DAIGAKU FU
356999CN3	706571JACT	MEDICAL CENTER
357850CN7		
357542CN0		
358946CN2	731425CTRT	JACKSON-MADISON CO GEN HOSP
358125CN3	C4200422	ROUEN/RAD.CENT.ECHO/SCAN/RDJ.
358486CN9		
358633CN6	QV0027	ZAIDANHOUSINTSUYA
358634CN4		
358124CN6		
359071CN8	727842CTRT	FLORIDA CANCER CENTER
358673CN2	414649LSCT5	AURORA MEDICAL GROUP
362228CN9		

359675CN6	512341RRCT2	ROUND ROCK HOSPITAL
358869CN6		
358702CN9		
359082CN5	205939SVOPLS	TRIMEDX
359159CN1	906774PRO	VA MED CTR IRON MOUNTAIN MI
359103CN9	704384PRO16	NOVANT HEALTH INC
359239CN1	954538LS16	MEMORIAL HOSPITAL
359497CN5	NO1016CT05	RIKSHOSPITALET HF
359494CN2		
359426CN4	516222CTRT	NASSAU RADIOLOGICAL GROUP
359313CN4	972981LSPRO	TEXAS HEALTH SYSTEM
359168CN2	651255RT	US ONCOLOGY
359427CN2	ACB11515RT	U OF TX MD ANDERSON CANCER CTR
359480CN1		
360160CN6	914681LSRT	WHITE PLAINS HOSP
359587CN3	2530246	LAWRENCE HOSPITAL
360316CN4	662234LS16P	BMH NORTH MISSISSIPPI
359732CN5		
359775CN4	405271RT1	OKLAHOMA UNIV MEDICAL CENTER
359879CN4		
360059CN0	972596LSRT	US ONCOLOGY
359951CN1	928779CT2	FlagStaff Medical Center
360060CN8	615338LS16	HENDERSONVILLE HOSPITAL/HTI
360158CN0		
360159CN8	815226PRO16	ST ANTHONYS HOSPITAL
360231CN5	SA1054CT01	KING FAHD MEDICAL
360723CN1		
361716CN4	413447RTP	BERKSHIRE MEDICAL CENTER
360494CN9	2516499	KAISER PERMENTE
360709CN0	217223BCT4	BLESSING HOSPITAL
360726CN4		
360413CN9		
360495CN6	937208CT1	NOVARE
360579CN7		
361007CN8		

361749CN5		
361293CN4		
362163CN8		
361715CN6	239436LS16	NAPLES COMMUNITY
361754CN5		
362705CN6		
362676CN9		
361841CN0	913596ONC16	PROVIDENCE MEDICA
362407CN9		
362288CN3		
362411CN1		
362758CN5		

Purpose

The spit detection circuit in the PPC board 2351200-11 rev 001 is too sensitive at high kV. At certain techniques, the natural ripple in the kV could cause false and repetitive spit detection leading to scan aborts due to too many spits. This is a risk for the patient. The scan aborts due to false spits could also lead to replacement of a tube that is not defective. This defect in PPC board 2351200-11 is a regression. PPC boards prior to 2351200-11 do not present this problem and the problem has been corrected in later board designs including 2407577. Please perform this FMIs per procedure below.

Related FMIs

FMI 25351 – LightSpeed Pro & RT (100kW 80kW & RT) Program M4 FMI

If this FMI is not performed before FMI25348 is implemented, you will have to follow the Appendix A for Flash Download Procedure that is very cumbersome and you may have to perform this step again when the FMI25351 will be install.

Furnished Materials

Qty	Part #	Description
1	2407577	PROGRAMMED KV CTRL P6.22BF2
1	5132113-100	FMI Doc.

Tools Required

Qty	Part #	Description	SW Version
1	5132113-100	FMI Doc.	N/A
	2345332	HIGH VOLTAGE MEASURING DIVIDER WITH ROTATING HANDLE	N/A
	OR		
	2375192 (alt)	HIGT VOLTAGE MEASURING DIVIDER COMPLETE PACKAGE	
2			

Aditonal Tools for HV Tank Feedback Resistor Verification.

Qty	Part #	Description	SW Version
1	N/A	Oscilloscope w/ 10x probe	N/A
1	N/A	Tube Hoist	N/A

Personnel Requirements

3 hours labor and 1 hours of travel

Procedure

The procedure below is a copy from the LightSpeed 5.X Service Information CD P/N 5114047-200. The procedure is located under Parts Replacement -> Gantry -> High Voltage section. For any clarification on this procedure please refer to this section.

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	30 mins	2 hrs

2 Overview

To replace JEDI PPC board:

- Save JEDI runtime parameters, if possible
- Replace PPC board
- Restore JEDI runtime parameters, if previously saved
- Check functionality of systems (see Finalization Section)
- Recalibrate (see Finalization Section)

3 Preliminary Requirements

3.4 Safety



⚠ WARNING

POTENTIAL FOR SHOCK.

VOLTAGE MAY BE PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS.

3.5 Required Conditions

Condition	Reference
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Before beginning this procedure, please read the gantry safety information.

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4 Procedure

4.1 PPC Board Removal

1. If possible, save JEDI runtime parameters (from JEDI to console) tool is located on Service desktop under Diagnostics\Generator Tools – JEDI. If this is not done JEDI history will be lost and calibration will take longer.



POTENTIAL FOR SHOCK.

VOLTAGE MAY BE PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS.

2. Remove power to the gantry using proper Lockout / Tagout procedures.
3. Remove gantry right side cover.
4. Turn OFF all the three switches (**AXIAL DRIVE ENABLE, HVDC ENABLE, 120 VAC**) on the Service Switch Panel.
5. Position the inverter assembly at 2 o'clock.



NOTICE

Do not drop screws into inverter. If you do, they must be removed or damage to the inverter will occur.

6. Remove largest inverter cover.
7. Attach a protective ESD wrist strap.
8. Disconnect wires and remove the PPC board by removing the seven screws (hex flange head) with a 5.5 mm socket.

4.2 PPC Board Installation

1. Continue using the ESD wrist strap.
2. Insert new PPC board and re-connect all wires.
3. Replace the seven screws (hex flange head).
4. Replace inverter cover.

5 Finalization

1. Ensure that proper torque specifications are followed for all fasteners.

Before running Flash Download please verify the rev. of the software by

1. Open a shell.
2. At the prompt enter the following command

```
showprods | grep Helios_OC ENTER
```

If the revision of software is other than 44.6 build please follow the instructions in Appendix A below before performing the Flash Download. If the rev. of software is equal to 44.6 build (this software was installed via FMI25351) proceed with the step 2 below.

2. Run Flash Download Tool to synchronize PPC firmware with Console software.
3. Restore JEDI Runtime Parameters, if you saved JEDI Runtime Parameters prior to replacing the PPC board. Location Service Desktop \ Diagnostics\Generator Tools – JEDI.
4. Run Meter Verification. Location Service Desktop \ Diagnostics\Generator Tools – JEDI.
5. Perform HV Tank Feedback Resistor Verification (see Appendix B for this procedure)
6. Run Filament Calibration. Location Service Desktop \ Diagnostics\Generator Tools – JEDI

Appendix A

1. Jedi Manual Flash download Procedure

Case 1: wrong database

- Open a UNIX shell
 - Type `cd /usr/g/GEfirm <ENTER>` to go to the GEfirm directory
- ```
{ctuser@qxis1}[4] cd /usr/g/GEfirm
{ctuser@qxis1}[5] sslInstall -jedi -char
```

Note: Below is a typical message during the flash donload.

Init Ecomm...

Attached to service service\_sw\_001..

Download JEDI Files....

Performing common ss char download..

Getting Database File stats for [/usr/g/ss\_fw/jedi/database.mx]

Database File Size = 14408

>>>> DOWNLOAD OK!! <<<<..

Performing common ss char download..

Reading/Prepping Char File [] for Download

errno: 2

In rdCharTable: No such file or directory

Error - rdCharTable read -1 bytes

**NOTE: The Following messages during the Jedi Manual download are normal -> errno: 2 and Error - rdCharTable read -1 bytes.**

**Pink window displaying “ One or more of the controllers (TGP/ORP/CCB/DCB/TCB/JEDI) or a system disk contains a missing or corrupt file. Please run flash download from Service Desktops tool to correct this problem”**

- Run Flash download tool **on the Utilities tab on the Service Desktop tool** to finish updating the JEDI software. **Continue Finalization at step 5.3 on previous page.**

## Appendix B

### *HV Tank Feedback Resistor Verification (Pro16, RT)*

#### 1 Personnel Requirements

| Required Persons | Preliminary Reqs | Procedure | Finalization   |
|------------------|------------------|-----------|----------------|
| 1                | Not Applicable   | 60 mins   | Not Applicable |

#### 2 Overview

This procedure is performed to ensure that the system is properly controlling scan parameters. By using the bleeder, verification is made on the secondary side of the HV tank and can be compared to selected scan parameters.

|               |               |
|---------------|---------------|
| Last Revised: | November 2004 |
|---------------|---------------|

#### 3 Preliminary Requirements

##### 3.1 Tools and Test Equipment

| Item                      | Qty | Part#   |
|---------------------------|-----|---------|
| Oscilloscope w/ 10x probe | 1   | -       |
| Tube Hoist                | 1   | -       |
| High Voltage bleeder kit  | 1   | 2345332 |

##### 3.2 Safety



##### **⚠ WARNING**

POTENTIAL FOR SHOCK.

VOLTAGE MAY BE PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS.

## 4 Procedure



### **WARNING**

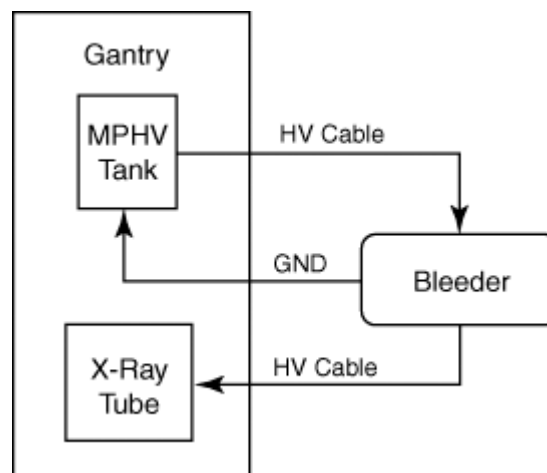
POTENTIAL FOR SHOCK.

VOLTAGE MAY BE PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS.

1. Remove gantry power. Be certain to follow appropriate Lockout/Tagout procedures.
2. Remove gantry right side cover.
3. Turn OFF all 3 switches (AXIAL DRIVE ENABLE, HVDC ENABLE, 120 VAC) on the Service Switch Panel.
4. Remove and set aside the top and front gantry covers.
5. Rotate the gantry such that HV Tank assembly is in the 3 o'clock position.
6. Lock the gantry in position, using the rotational lock. Ensure that gantry rotation is locked by attempting to rotate the gantry by hand.
7. Free the HV cable.
  - a. Carefully cut tie-wraps securing HV cable. Note HV cable routing.
  - b. Loosen the cable's locking ring with a spanner wrench.
  - c. Pull cable terminal stick out of its receptacle on HV tank.
8. Install the HV bleeder between the tank and the tube. Use the sling/hoist to suspend the bleeder. See Illustration 1.

**Illustration 1: Bleeder Connection Block Diagram**



### **NOTICE**

Ensure that you attach a ground cable from the bleeder to the HVT. Incorrect bleeder setup can result in damage to the HV Subsystem.

9. Turn ON the 120VAC and HVDC ENABLE on the Service Switch Panel.

10. Wait about 2 minutes for the gantry display to reset. No flashing lights on the gantry display should indicate this.

11. Clear the E-Stop condition by pressing the DRIVES RESET button on the Service



Switch Panel or on the gantry control panel.

12. Setup the oscilloscope:

- a. To minimize noise and obtain a common ground, you should plug the oscilloscope into either the gantry or the console.
- b. Use 10x probes
- c. Connect channel one probe to the positive (red) output of the bleeder and ground to the negative (black) output of the bleeder.
- d. Set horizontal scale = 200 ms, vertical scale = 10V/div
- e. Trigger channel one, negative slope, DC couple, trigger mode normal.

13. Launch [kV and mA Tool].

14. Verify the following scan parameters:

- X-Ray test type = manual
- Gantry = Disabled
- Exposure time = 1s
- No of scans = 1
- Focal Spot = Large
- kV = 80kV
- mA = 50mA

15. Press RUN and take scan.

16. Measure the channel 1 mean from just inside the front edge to just inside the back edge of the waveform, using cursors on the oscilloscope.

17. Repeat the measurements for 100, 120, and 140kV scans.

18. Verify internal scan timer:

- a. Change trigger slope on oscilloscope to positive. Adjust trigger level to TTL if necessary.
- b. Take the following scan:
  - X-Ray test type = manual
  - Gantry = Disabled
  - Exposure time = 1s
  - No of scans = 1
  - Focal Spot = Large
  - kV = 100kV
  - mA = 40mA

19. Measure the output

Using cursors on the oscilloscope, measure the scan time (Dt) from just inside the front edge to just inside the back edge of the waveform.

**NOTICE**

**Incorrectly sealed and secured HV cable can result in damage to the tank or other parts of the gantry.**

20. Reconnect HV cable from tube to tank. Make sure you overfill the receptacle.  
Reference: Securing HV Cable (Pro 16).

## 5 Finalization

1. The scale factor for your kV measurements is 10kV/volt (e.g. 8V => 80kV).
2. To pass the kV check, all three of the following conditions must be met:
  - The scope reading must be within 3% of the desired kV.
  - The kV mA Tool screen reading must be within 3% of the desired kV.
  - Both measurements (scope and screen) must be within 2% of each other.
3. The internal scan timer measurement should be within +/- 4%.

Record the results on the HHS data sheet and / or take appropriate service action.

NOTES: